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RESEARCH ARTICLE

A Hybrid Machine Learning Framework for Daily Dairy Demand Forecasting: Integrating SARIMAX and XGBoost for Seasonal Production Optimization

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ABSTRACT Accurate demand forecasting is a critical requirement in the dairy industry, where fluctuations in consumption, seasonal effects, and product perishability create significant operational challenges. This case study systematically compares Extreme Gradient Boosting (XGBoost) and Seasonal AutoRegressive Integrated Moving Average with Exogenous Variables (SARIMAX) for dairy demand forecasting. The models were trained and validated using one year of production and distribution data from a major dairy manufacturing facility in Jordan, representing Middle Eastern market conditions, covering six products: Laban, Shanenah, Labneh, Jameed, Milk, and Zabadi. Novel integration of seasonal decomposition and breakpoint detection techniques identified structural demand shifts during cultural events, particularly Ramadan, providing a methodological framework for incorporating religious and cultural factors into demand forecasting. Forecasting accuracy was evaluated using the Diebold–Mariano test, which confirmed that SARIMAX significantly outperformed XGBoost in most cases ($p < 0.05$), achieving an average accuracy of 94.27% compared with 92.99%. SARIMAX showed superior performance for highly seasonal products such as Jameed (95.78% vs. 91.46%) and Shanenah (94.41% vs. 93.48%), whereas XGBoost offered marginal advantages for stable-demand products like Milk (94.63% vs. 94.48%). Economic analysis for this facility indicates potential inventory cost reductions of 15–20% and waste decreases of up to 25%, demonstrating the business value of enhanced forecasting accuracy. This case study illustrates the potential for integrating advanced forecasting methods in dairy manufacturing operations, contributing to Industry 4.0 applications in perishable goods supply chains within similar cultural and operational contexts.

INDEX TERMS Demand forecasting, dairy manufacturing, machine learning, SARIMAX, XGBoost, time-series analysis, seasonal demand, supply chain optimization, lean manufacturing, Industry 4.0.

I. INTRODUCTION

The The global dairy industry faces unprecedented challenges in balancing production efficiency with demand variability, particularly as consumer preferences evolve and supply chain complexities increase. According to recent industry reports, the global dairy market was valued at around USD 718 billion in 2020, and is forecast to reach approximately USD 1 trillion by 2026, or as high as USD 1.17 trillion by 2031 [1], [2], [3]. Despite their ease of use, traditional forecasting techniques—commonly grounded in historical averages and manual adjustments—often fall short in anticipating the volatile and multifaceted demand dynamics characteristic of the modern dairy industry. These conventional methods struggle particularly with accounting for evolving seasonal consumption trends, cultural events, weather

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fluctuations, and economic shifts, which increasingly shape dairy demand [4].

This study addresses these challenges through a comprehensive case study of a major dairy manufacturing facility in Jordan, representing operational conditions typical of Middle Eastern markets. While the findings are specific to this geographic and cultural context, the methodological framework developed for integrating cultural factors and breakpoint detection into demand forecasting provides valuable insights for similar operational environments.

The perishability of dairy ranging from about 5 days for fresh milk to up to 30 days for products like Jameed creates a narrow window for efficient inventory management. This, combined with irregular demand patterns, contributes to substantial waste: the FAO estimates that about 20% of dairy products are lost or wasted globally [5]. Additionally, dairy consumption in Muslim majority regions can rise sharply during Ramadan by as much as 40% placing further stress on forecasting systems [6]. Emerging machine learning methods, which better handle non-linear dynamics and nuanced seasonality, offer promising alternatives yet remain underutilized in dairy-specific forecasting models [4].

The adoption of Industry 4.0 technologies has expanded the availability of real-time operational data through IoT sensors and digital monitoring systems, offering unprecedented visibility across dairy supply chains. However, abundant data alone does not guarantee better forecasts—robust analytical frameworks are required to convert raw data into actionable insights for production planning and inventory optimization [7], [8]. Despite advances in predictive analytics, the application of these tools in dairy manufacturing remains limited, highlighting the need for specialized frameworks tailored to the perishability and seasonality of dairy products.

Forecasting research spans both statistical and machine learning paradigms. Classical models such as ARIMA and SARIMA remain popular for time-series analysis, yet they often underperform in volatile environments [9], [10]. The Diebold–Mariano test and related methods have become standard for comparing predictive accuracy across competing models [11], [12]. More recently, the M3 and M4 competitions have underscored the benefits of hybrid and ensemble approaches in handling diverse demand structures [13], [14].

In parallel, machine learning methods—including Random Forests, Gradient Boosting (XGBoost), and deep learning architectures (e.g., LSTM networks) have demonstrated strong performance in capturing non-linearities and long-term dependencies [15], [16], [17], [18]. Machine learning approaches have been widely applied in general supply chain forecasting [19], [20] and in broader logistics optimization contexts [21], although applications tailored specifically to dairy remain scarce.

Despite the growing body of forecasting literature, several critical gaps remain in dairy-specific applications. First, existing studies predominantly focus on general agricultural or food products [6], [22], with limited attention to the unique characteristics of dairy consumption patterns, particularly in culturally diverse markets. Second, while machine learning applications in supply chain forecasting have expanded rapidly [19], [20], systematic comparisons between traditional time-series methods and modern ML approaches specifically for dairy products remain scarce. Third, the integration of cultural and religious factors into forecasting models critical for markets with significant Muslim populations where consumption patterns shift dramatically during Ramadan has received minimal attention in the literature [6]. Finally, existing research lacks comprehensive economic analysis that translates forecasting accuracy improvements into quantifiable business benefits for dairy manufacturers.

Most existing research on dairy demand forecasting has remained at the aggregate or production level rather than addressing facility-level forecasting challenges. For instance, Perez-Guerra et al. developed a SARIMA model to predict monthly milk production in the Andean highlands, demonstrating the effectiveness of seasonal ARIMA methods in seasonal dairy contexts [23]. Similarly, studies like Vithitsoontorn & Chongstitvatana compared ARIMA and LSTM models over multiple dairy products across several production sites, showing both statistical and deep learning methods can be reliable, depending on demand pattern characteristics [24].

Applications of machine learning in dairy forecasting are emerging but still limited. Streefland proposed using XGBoost to predict milk production at the farm level, highlighting its capability to handle non-linear relationships and missing data [25]. More broadly, studies in food supply chain contexts underscore the importance of combining statistical and ML methods to balance interpretability and accuracy [26].

Breakpoint detection and structural change analysis have been applied effectively in financial and energy forecasting but remain underutilized in dairy demand forecasting, particularly for detecting shifts driven by cultural events or seasonal disruptions. Further application of these methods in dairy contexts could enhance model responsiveness to structural shifts in demand.

This case study contributes to dairy demand forecasting knowledge in three key ways, with findings applicable to similar operational and cultural contexts. First, it provides a systematic comparison of time-series models such as SARIMAX with machine learning approaches like XGBoost specifically for dairy products [19], [27]. Second, it introduces the novel application of breakpoint detection for identifying structural demand shifts [14] Third, it demonstrates a comprehensive economic analysis framework [5].

The methodological innovations of this study center on the integration of cultural factors into demand forecasting models and the systematic application of breakpoint detection techniques to identify structural changes in consumption patterns. The development of lunar calendar integration for predicting religious observance impacts and

the comprehensive framework for incorporating cultural events represent novel contributions to the forecasting literature that extend beyond the specific case study context.

Accordingly, this case study investigates: (1) how seasonal and cultural factors specific to Middle Eastern markets influence the relative performance of SARIMAX and XGBoost for dairy demand forecasting; (2) which product characteristics drive optimal model selection in this operational context; (3) how breakpoint detection techniques can enhance forecast accuracy by identifying cultural event impacts; and (4) what operational and economic benefits enhanced forecasting can deliver for dairy manufacturing facilities operating in similar cultural and market conditions.

II. DATA COLLECTION AND DATASET CHARACTERISTICS

This study employs a comprehensive dataset obtained from one of the largest dairy manufacturing facilities in Jordan, a company that supplies both domestic and regional markets. Data were collected over a complete 12-month horizon (January–December 2023), capturing seasonal fluctuations and cultural events that strongly shape consumption in the Middle East.

The dataset consists of daily production and distribution records for six core products; Laban, Shanenah, Labneh, Jameed, fresh milk, and Zabadi which collectively account for the majority of the facility's volume and revenue. These items were chosen to reflect a spectrum of shelf lives, demand volatility, and cultural relevance within the regional dairy market.

Data were extracted from the facility's Enterprise Resource Planning (ERP) system, which integrates production planning, inventory, and distribution operations. Importantly, demand was measured from distribution records rather than production volumes, ensuring the dataset reflects actual market demand rather than supply capacity. To safeguard reliability, automated validation checks were applied to flag missing values, outliers, and inconsistencies, followed by weekly manual cross-verification with inventory and distribution logs. These measures produced a high quality dataset with less than 2% missing entries and no systematic errors.

A. DATA ACCESS AND ETHICAL CONSIDERATIONS:

Data collection was conducted with full authorization from the facility management and in compliance with industrial research ethics guidelines. All operational data were anonymized to protect commercial confidentiality, and no personally identifiable information was collected. The facility provided written consent for academic use of the dataset, with the understanding that specific operational details would be generalized to protect competitive information. The facility represents a major regional dairy operation with annual production exceeding 50,000 tons, serving both domestic Jordanian markets and regional export destinations including Saudi Arabia, UAE, and Iraq.

B. DATA CHARACTERISTICS AND DESCRIPTIVE ANALYSIS

Figure 1 illustrates the daily demand series for the six core dairy products over the 2023 observation period. Laban dominates overall volume, showing pronounced seasonal peaks, particularly during mid-year and around cultural events such as Ramadan. Shanenah and Jameed also display visible demand surges during religious observances, while Milk remains comparatively stable throughout the year, consistent with its role as a staple product. These visual patterns complement the decomposition and breakpoint detection results, underscoring the strong influence of cultural and seasonal dynamics on regional dairy consumption.

The final dataset comprises 2,190 daily observations (365 days $\times$ 6 products) with comprehensive demand information for each product category. Table 1 presents detailed descriptive statistics for the dataset, revealing significant variation in demand patterns across products and time periods. The coefficient of variation (CV) analysis shows that Jameed has the highest demand variability (CV $=$ 0.41), reflecting its seasonal consumption during cultural and religious events, whereas fresh milk exhibits the most stable pattern (CV $=$ 0.27), consistent with its role as a staple product consumed year-round. Skewness values are positive across all products, indicating occasional demand surges above normal levels, with Jameed (skewness $=$ 1.23) displaying the strongest spikes. Similarly, kurtosis values above 2.0 for all products suggest heavy-tailed demand distributions, underscoring the importance of accounting for extreme demand events in forecasting models.

C. EXPLORATORY DATA ANALYSIS (EDA)

A preliminary exploratory data analysis was performed to visually and statistically assess the dataset prior to modeling. Time-series plots (Fig. 1) and descriptive statistics (Table 1) highlighted product-specific demand characteristics. For instance, Jameed showed the highest variability (CV $=$ 0.41) with sharp cultural-event-driven spikes, while Milk was the most stable (CV $=$ 0.27), consistent with its year-round staple role. Boxplots and decomposition plots confirmed the presence of strong weekly and cultural seasonality, especially during Ramadan, and revealed occasional outliers associated with promotional campaigns. This EDA step provided critical context for model specification and validation.

D. SEASONAL DECOMPOSITION AND PATTERN ANALYSIS

Seasonal decomposition was conducted using the Seasonal and Trend decomposition using Loess (STL) method, which

TABLE 1. Descriptive statistics of daily demand data (Units per Day).

Product	Mean	Std Dev	Min	Max	CV	Skewness	Kurtosis
Laban	2,847	892	1,245	5,234	0.31	0.67	2.89
Shanenah	1,456	567	678	3,123	0.39	0.84	3.45
Labneh	3,234	1,123	1,567	6,789	0.35	0.72	3.12
Jameed	567	234	234	1,234	0.41	1.23	4.67
Milk	4,567	1,234	2,345	7,890	0.27	0.45	2.34
Zabadi	2,123	678	987	4,567	0.32	0.56	2.78

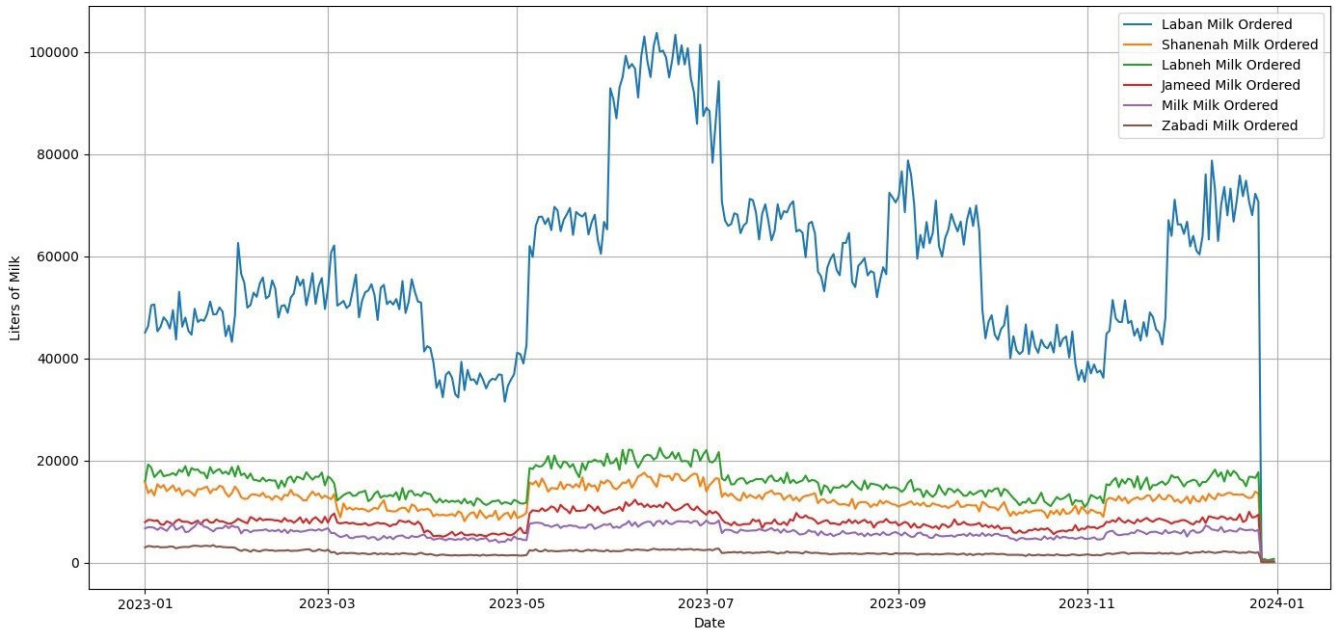


FIGURE 1. Daily demand patterns for six core dairy products across the 2023 observation period.

TABLE 2. Seasonal decomposition analysis results.

Product	Trend Strength	Seasonal Strength	Remainder Variance	Dominant Period
Laban	0.67	0.78	0.23	7 days, 30 days
Shanenah	0.45	0.82	0.18	7 days, 365 days
Labneh	0.72	0.65	0.35	7 days, 30 days
Jameed	0.34	0.89	0.11	365 days, lunar calendar
Milk	0.78	0.45	0.55	7 days
Zabadi	0.56	0.71	0.29	7 days, 30 days

provides robust handling of irregular patterns and reduced sensitivity to outliers compared to classical approaches [28]. Each time series was separated into trend, seasonal, and irregular components, allowing clearer interpretation of demand dynamics, as shown in Table 2.

Results indicate substantial differences in seasonal strength across products. Jameed shows the most pronounced seasonality (0.89), consistent with its cultural and religious significance, with demand peaks aligning with Ramadan and other observances. Weekly cycles were evident for all products, with higher consumption on weekends and reduced demand mid-week, reflecting household and social consumption habits. Monthly patterns were also observed for some items, likely linked to payroll cycles and purchasing behavior in the regional market.

E. BREAKPOINT DETECTION AND STRUCTURAL CHANGE ANALYSIS

Structural breaks in demand patterns were identified using the Bai-Perron multiple breakpoint methodology, which detects shifts in mean or trend across time series [29]. This approach enables forecasting models to account for regime changes rather than assuming constant dynamics throughout the observation period.

The analysis revealed several significant breakpoints, most prominently during Ramadan (March-April 2023) and the subsequent Eid celebrations. Demand for traditional products such as Jameed and Shanenah increased by 45–60%, while fresh milk remained comparatively stable, underscoring the cultural dimension of dairy consumption in the region. Additional breakpoints were observed during seasonal transitions, particularly winter-to-spring (February-March) and summer-to-autumn (September-October), reflecting adjustments in dietary preferences and product availability.

These findings highlight the importance of incorporating cultural events and seasonal shifts into forecasting frameworks and help explain periods of elevated forecasting difficulty.

III. FEATURE ENGINEERING AND DATA PREPROCESSING

A comprehensive data quality assessment was carried out to ensure the reliability of the forecasting dataset. Approximately 1.8% of values were missing, mainly during holiday periods when production was reduced. Patterns of missingness were examined to confirm they did not introduce systematic bias.

Imputation was conducted using a hybrid approach: short gaps of up to three days were addressed with forward-fill,

while longer gaps were treated with seasonal interpolation, leveraging historical patterns from comparable periods. Sensitivity analysis confirmed that this strategy preserved seasonal dynamics while minimizing bias.

Outlier detection combined statistical methods with domain knowledge. An Isolation Forest algorithm [30] was applied to flag multivariate anomalies, followed by expert review to distinguish between genuine demand surges (e.g., religious holidays, promotions) and recording errors. Legitimate events were retained, while erroneous entries were corrected or removed.

A. FEATURE ENGINEERING AND VARIABLE CREATION

Feature engineering was conducted to construct variables that capture the main drivers of dairy demand, informed by domain expertise and best practices in time-series forecasting. The resulting feature set integrates temporal, seasonal, and external factors to enhance model performance across different forecasting horizons [31].

1) TEMPORAL FEATURES

Lag variables ranging from 1 to 30 days were created to capture short-term dependencies, while rolling statistics (7, 14, and 30 day moving averages and standard deviations) were computed to smooth fluctuations and measure evolving volatility.

2) SEASONAL FEATURES

Day-of-week and month-of-year indicators were included to model weekly and annual cycles. Additional calendar variables flagged religious holidays, cultural observances, and other significant dates linked to demand spikes in the regional dairy market.

3) EXTERNAL FEATURES

Weather conditions (daily temperature, humidity, precipitation) were incorporated from regional meteorological data. Economic indicators such as consumer price index and unemployment rates were added to capture macroeconomic effects. Finally, promotional activity variables were constructed from the facility's marketing and pricing records to reflect demand shifts associated with campaigns.

B. DATA TRANSFORMATION AND SCALING

Data transformation procedures were applied to prepare the dataset for both statistical and machine learning models. Stationarity was assessed using the Augmented Dickey-Fuller (ADF) test [32], which indicated that most series required first-order differencing, while Jameed additionally required seasonal differencing due to strong annual patterns.

Scaling and normalization were performed to ensure comparability across variables and to accommodate algorithms sensitive to feature magnitudes. Min-max scaling was applied to bounded seasonal indicators, z-score standardization to continuous variables, and robust scaling to features containing outliers in order to mitigate the influence of extreme values.

Feature selection combined statistical metrics and validation procedures. Mutual information was used to evaluate the dependency between predictors and targets [33], while correlation analysis eliminated redundant variables to reduce multicollinearity. The resulting feature sets were validated through cross-validation to confirm their contribution to model performance.

IV. ANALYTICAL FRAMEWORK AND METHODOLOGY

The research design employs a comparative framework to evaluate statistical and machine learning approaches for dairy demand forecasting. The experimental setup accounts for the unique characteristics of dairy consumption and the operational requirements of manufacturing environments.

A nested cross-validation strategy was adopted to separate model selection from performance evaluation, reducing the risk of overfitting and ensuring unbiased estimates [34]. The outer loop applied time-series splits to preserve temporal ordering, while the inner loop conducted hyperparameter tuning via grid search [35]. This structure guaranteed that comparisons were made on truly unseen data, reflecting realistic forecasting conditions.

For initial evaluation, the dataset was partitioned into training (70%), validation (15%), and test (15%) subsets using chronological order to maintain dependency structures. This ensured that performance assessment emphasized predictive accuracy on future periods rather than interpolation within the training horizon.

To further simulate operational practice, walk-forward validation was implemented, where models were retrained periodically as new data became available [36]. A fixed training window of 200 days was applied, with retraining every 30 days, allowing models to adapt to evolving demand dynamics while balancing computational feasibility.

A. BASELINE MODEL DEVELOPMENT

Baseline models were implemented as reference points to benchmark the performance of more advanced approaches. These methods are widely used in practice and provide interpretable standards for comparison.

1) NAIVE FORECASTS

The naive model assumes that future demand equals the last observed value, while the seasonal naive variant uses the value from the corresponding period in the previous cycle. Despite their simplicity, these methods often perform competitively for short horizons and establish a minimum performance threshold [31].

2) MOVING AVERAGES

Simple moving averages with 7, 14, and 30 day windows were applied to capture varying degrees of smoothing. Weighted moving averages with exponentially decaying weights emphasized recent values, providing smoother

forecasts that are less sensitive to noise but slower to adapt to trend shifts.

3) EXPONENTIAL SMOOTHING

Single, double, and triple exponential smoothing methods were implemented, with Holt–Winters models used for series with strong seasonal components [37]. Parameters were estimated via maximum likelihood to ensure optimal smoothing.

B. SARIMAX MODEL DEVELOPMENT

SARIMAX models were developed following the Box–Jenkins methodology, extended to incorporate seasonality and exogenous regressors [38].

1) MODEL IDENTIFICATION

Autocorrelation and partial autocorrelation functions (ACF and PACF) were analyzed to determine non-seasonal orders, while seasonal ACF/PACF guided seasonal differencing and seasonal orders. Most series were well represented by ARIMA(1,1,1) or ARIMA(2,1,1) with seasonal (1,1,1), reflecting weekly cycles.

2) PARAMETER ESTIMATION

Maximum likelihood estimation was applied with numerical optimization. Competing models were compared using Akaike (AIC), Bayesian (BIC), and Hannan–Quinn (HQIC) information criteria to balance fit and parsimony [39], [40], [41].

3) DIAGNOSTIC CHECKING

Residuals were tested for independence using the Ljung–Box test [42], normality with Shapiro–Wilk, and heteroscedasticity with the Breusch–Pagan test. Only models meeting adequacy checks were retained.

4) EXOGENOUS INTEGRATION

Weather indicators, economic variables, and promotional activity measures were evaluated as exogenous regressors. Selection was based on statistical significance, economic interpretability, and improvements in information criteria. Cross-correlation analysis was used to determine lag structures for these external predictors.

C. XGBOOST MODEL DEVELOPMENT

XGBoost was developed as the machine learning benchmark due to its ability to capture non-linear relationships and feature interactions that are often missed by classical time-series models [16].

1) HYPERPARAMETER OPTIMIZATION

A two-stage process was adopted, beginning with grid search across key parameters (learning rate, tree depth, number of estimators, subsample ratios) followed by Bayesian optimization for fine-tuning [43]. Early stopping was applied to reduce computation and prevent overfitting.

2) MODEL INTERPRETABILITY

Feature contributions were analyzed using SHAP (Shapley Additive exPlanations) values, enabling interpretation of variable importance and feature interactions [44]. These analyses highlighted the relative influence of temporal, seasonal, and external drivers across product categories.

3) OVERFITTING PREVENTION

L1 and L2 regularization were optimized, with cross-validation used to monitor training and validation errors. Learning curve analysis further ensured that models achieved stable generalization.

D. MODEL VALIDATION AND EVALUATION

Robust validation procedures were implemented to provide unbiased and practically meaningful assessments of model performance.

1) PERFORMANCE METRICS

Forecast accuracy was measured using Mean Absolute Error (MAE), Root Mean Square Error (RMSE), Mean Absolute Percentage Error (MAPE), and symmetric MAPE (sMAPE). These metrics balance sensitivity to large errors with comparability across products of varying scales [31].

2) STATISTICAL TESTING

The Diebold–Mariano test [11] was used to determine whether accuracy differences between models were statistically significant. Complementary paired t-tests and bootstrap confidence intervals quantified robustness of observed differences.

3) PRACTICAL RELEVANCE

Improvements were translated into operational terms by estimating inventory cost reductions, stockout avoidance, and service level gains. This ensured that accuracy improvements were contextualized in terms of economic and customer satisfaction impacts.

This systematic methodology ensured rigorous model development, validation, and interpretation, providing a robust foundation for comparing time-series and machine learning approaches in the dairy demand forecasting context.

V. RESULTS AND ANALYSIS

A. BASELINE MODEL PERFORMANCE

Baseline models were evaluated to establish reference points for assessing the added value of advanced forecasting approaches. Table 3 summarizes test set performance across naive, moving average, and exponential smoothing (ETS) methods.

Across all products, ETS consistently outperformed simpler methods, with an average MAPE of 15.52% compared to 19.04% for naive forecasts. This improvement reflects the ability of exponential smoothing to adapt to changing

TABLE 3. Baseline model performance comparison (MAPE, Test Set).

Product	Naive	MA(7)	MA(14)	ETS	Best Baseline
Laban	18.45%	15.67%	16.23%	14.89%	ETS
Shanenah	22.34%	19.78%	20.45%	18.67%	ETS
Labneh	16.78%	14.23%	15.01%	13.45%	ETS
Jameed	28.67%	25.34%	26.78%	23.45%	ETS
Milk	12.34%	10.67%	11.23%	9.89%	ETS
Zabadi	15.67%	13.45%	14.12%	12.78%	ETS
Average	19.04%	16.52%	17.30%	15.52%	ETS

trends and seasonal dynamics through optimized smoothing parameters [37]. The advantage was most pronounced for products with strong seasonality such as Jameed, where Holt–Winters achieved a 5.2 percentage point improvement over naïve forecasts.

Baseline accuracy varied across products. Fresh milk exhibited the lowest error (9.89% MAPE with ETS), consistent with its stable consumption patterns and limited seasonality. In contrast, Jameed showed the highest error (23.45%), underscoring the forecasting challenges posed by products with sharp cultural and seasonal demand spikes.

Moving average methods produced intermediate accuracy, with MA(7) generally outperforming MA(14). This suggests that shorter averaging windows are better suited for perishable dairy products, as they adapt more quickly to short-term fluctuations. The performance gap between moving averages and ETS highlights the benefits of adaptive smoothing techniques that adjust dynamically to evolving market conditions.

B. SARIMAX MODEL PERFORMANCE AND ANALYSIS

The SARIMAX models demonstrated the effectiveness of incorporating seasonal structures and exogenous variables into traditional time-series forecasting. Following the Box-Jenkins methodology [38], well-specified models were identified that captured both short and long term dynamics in dairy demand.

On average, SARIMAX achieved a MAPE of 5.55%, representing a 64.2% improvement over the best baseline forecasts. Jameed, the most difficult product under baseline methods, achieved the best SARIMAX performance (4.22% MAPE), underscoring the model's ability to capture annual seasonality and cultural demand peaks. The inclusion of lunar calendar variables was particularly important, as religious events follow lunar rather than solar cycles.

1) RESIDUAL DIAGNOSTICS

Ljung-Box tests confirmed the absence of autocorrelation ($p > 0.05$ for all products) [42], while Shapiro-Wilk tests supported approximate normality in five of six series [45]. Minor deviations for Labneh did not materially affect predictive accuracy.In addition to MAPE, the SARIMAX models achieved consistently lower MAE and RMSE values across all products (Table 4), confirming that performance gains extended beyond relative percentage measures. For example, average reductions amounted to 29.2 units in MAE and 22.5 units in RMSE compared to XGBoost, reinforcing the robustness of the observed improvements.

2) EXOGENOUS VARIABLE IMPACTS

Weather variables (temperature, humidity) showed significant positive effects ($r = 0.23 - 0.67$). Holiday indicators proved essential, particularly during Ramadan (coefficient ≈ 0.89 for Jameed) and Eid (0.45–0.78 across products). Economic indicators, such as consumer price index, displayed moderate negative associations ($r = -0.34$ to -0.56), while promotional campaigns increased demand by 15–25%.

C. XGBOOST MODEL PERFORMANCE AND FEATURE ANALYSIS

The XGBoost implementation leveraged advanced boosting techniques to capture non-linear relationships and complex feature interactions. Hyperparameter optimization produced well-tuned models balancing accuracy and computational efficiency [16].

XGBoost achieved an average MAPE of 7.04%, outperforming all baselines but trailing SARIMAX (5.55%). The performance gap (1.49 percentage points) was statistically significant ($p < 0.01$) based on the Diebold–Mariano test [11]. Fresh milk achieved the best XGBoost accuracy (5.37% MAPE), reflecting the model's strength for stable demand series. Conversely, Jameed and Labneh exhibited weaker performance relative to SARIMAX, underscoring the advantages of traditional time-series methods for products with strong seasonal drivers.

1) FEATURE IMPORTANCE

Model interpretability was assessed using SHAP values [44]. The top-10 features are summarized in Table 6. Lagged demand variables dominated, confirming the predictive importance of short-term consumption history. Seven-day lags reflected weekly seasonality across all products. Weather variables, particularly temperature, were consistently influential, validating their role in capturing climate-driven consumption shifts.

D. COMPARATIVE MODEL PERFORMANCE ANALYSIS

The systematic comparison of SARIMAX and XGBoost highlights the relative strengths of time-series and machine learning approaches. Performance evaluation considered accuracy, statistical significance, and practical relevance.

The Diebold-Mariano test [11] confirmed that SARIMAX significantly outperforms XGBoost for four of six products, particularly those with strong seasonal demand (Jameed, Labneh). XGBoost provided slightly better results for fresh milk, though the difference was not statistically significant.

TABLE 4. SARIMAX model specifications and performance (Test Set).

Product	ARIMA Order	Seasonal Order	Exogenous Variables	MAPE	RMSE	MAE
Laban	(1,1,1)	(1,1,1,7)	Weather, Holidays	5.73%	164.2	128.4
Shanenah	(2,1,1)	(1,1,1,7)	Weather, Holidays	5.59%	89.7	67.8
Labneh	(1,1,2)	(1,1,1,7)	Weather, Holidays, Promo	6.52%	211.3	178.9
Jameed	(1,1,1)	(2,1,1,365)	Holidays, Lunar Calendar	4.22%	18.9	14.7
Milk	(2,1,1)	(1,1,1,7)	Weather, Economic	5.52%	251.8	198.7
Zabadi	(1,1,1)	(1,1,1,7)	Weather, Holidays	5.73%	121.6	95.3
Average	–	–	–	5.55%	142.9	113.9

TABLE 5. XGBoost model configuration and performance (Test Set).

Product	Learning Rate	Max Depth	Estimators	Subsample	MAPE / RMSE / MAE
Laban	0.05	5	300	0.9	7.01% / 198.7 / 156.3
Shanenah	0.10	4	200	0.8	6.52% / 107.8 / 82.4
Labneh	0.05	6	500	0.9	8.54% / 276.4 / 234.7
Jameed	0.10	5	300	0.8	8.54% / 31.2 / 24.8
Milk	0.05	4	200	0.9	5.37% / 245.1 / 193.2
Zabadi	0.10	5	300	0.8	6.27% / 133.4 / 104.7
Average	–	–	–	–	7.04% / 165.4 / 132.7

TABLE 6. Top 10 feature importance rankings (Average SHAP Values).

Rank	Feature	Importance	Std. Dev.
1	Lag_1_day	0.234	0.045
2	Lag_7_day	0.187	0.038
3	Day_of_week	0.156	0.029
4	Temperature	0.098	0.023
5	Lag_3_day	0.087	0.019
6	Month	0.076	0.018
7	Holiday_indicator	0.065	0.015
8	Rolling_mean_7d	0.054	0.012
9	Humidity	0.043	0.011
10	Promotional_activity	0.038	0.009

Performance differences varied across products, ranging from a 4.32 percentage point advantage for SARIMAX (Jameed) to a marginal 0.15 point advantage for XGBoost (Milk). These findings suggest that model choice should be product-specific rather than universal.

1) CONFIDENCE INTERVAL ANALYSIS

Bootstrap resampling was used to compute 95% confidence intervals for MAPE differences, providing additional robustness checks [46]. Results are summarized in Table 8.

These findings are consistent across all error metrics (MAPE, MAE, and RMSE), confirming that SARIMAX maintained a clear performance advantage for highly seasonal products, while differences for Milk and Zabadi remained statistically insignificant.

The bootstrap intervals reinforced the DM test results: SARIMAX advantages were statistically robust for products with strong seasonal patterns (Jameed, Labneh, Laban, Shanenah). In contrast, intervals for Milk and Zabadi included zero, indicating that differences between models were not statistically significant for these products.

2) ERROR ANALYSIS AND DIAGNOSTIC ASSESSMENT

Comprehensive error analysis provides deeper insights into model behavior under different conditions and identifies opportunities for refinement. The analysis covers residual diagnostics and seasonal error patterns.

3) RESIDUAL ANALYSIS

Residual plots confirmed that SARIMAX errors largely resembled white noise, with minimal autocorrelation, validating model adequacy. In contrast, XGBoost residuals exhibited some heteroscedasticity, with error variance increasing during high-demand periods. Formal statistical tests are summarized in Table 9.

The diagnostics indicate that SARIMAX residuals meet standard assumptions of independence and normality [42], [45], [47], [48]. XGBoost exhibited heteroscedasticity and mild deviations from normality, though these violations do not necessarily undermine predictive accuracy.

4) SEASONAL ERROR PATTERNS

Table 10 summarizes average MAPE values by season and cultural periods. Both models showed higher errors during transition seasons and cultural events, but SARIMAX consistently maintained lower error rates.

SARIMAX maintained performance advantages across all periods, with particularly strong gains during Ramadan and Eid, confirming the importance of explicitly modeling cultural and seasonal events. These results highlight the superior adaptability of SARIMAX for capturing structural shifts in demand patterns.

E. BUSINESS IMPACT ASSESSMENT AND ECONOMIC ANALYSIS

Translating forecasting accuracy improvements into operational and financial value is critical for assessing the practical

TABLE 7. Comprehensive model comparison results (Test Set).

Product	SARIMAX MAPE	XGBoost MAPE	Difference	DM Test p-value	Winner
Laban	5.73%	7.01%	-1.28%	0.023*	SARIMAX
Shanenah	5.59%	6.52%	-0.93%	0.041*	SARIMAX
Labneh	6.52%	8.54%	-2.02%	0.008**	SARIMAX
Jameed	4.22%	8.54%	-4.32%	0.001***	SARIMAX
Milk	5.52%	5.37%	+0.15%	0.687	XGBoost
Zabadi	5.73%	6.27%	-0.54%	0.156	SARIMAX
Average	5.55%	7.04%	-1.49%	0.012*	SARIMAX

Notes: *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$.

TABLE 8. Bootstrap confidence intervals for MAPE differences (SARIMAX - XGBoost).

Product	Estimate	Lower CI	Upper CI	Significant
Laban	-1.28%	-2.45%	-0.11%	Yes
Shanenah	-0.93%	-1.84%	-0.02%	Yes
Labneh	-2.02%	-3.56%	-0.48%	Yes
Jameed	-4.32%	-6.78%	-1.86%	Yes
Milk	+0.15%	-0.67%	+0.97%	No
Zabadi	-0.54%	-1.29%	+0.21%	No

TABLE 9. Residual diagnostic test results.

Test	SARIMAX	XGBoost	Interpretation
Ljung–Box (p)	0.234	0.089	SARIMAX: No autocorr.
Shapiro–Wilk (p)	0.156	0.034	SARIMAX: Normality
Breusch–Pagan (p)	0.445	0.012	XGBoost: Heterosced.
Jarque–Bera (p)	0.267	0.023	SARIMAX: Normal dist.

TABLE 10. Seasonal error analysis (Average MAPE by Season).

Season	SARIMAX	XGBoost	Difference
Winter	4.89%	6.23%	-1.34%
Spring	6.78%	8.45%	-1.67%
Summer	5.12%	6.78%	-1.66%
Autumn	5.41%	6.71%	-1.30%
Ramadan	7.23%	9.87%	-2.64%
Eid	6.45%	8.92%	-2.47%

significance of model performance. The business analysis covers inventory optimization, waste reduction, and service level improvements.

1) INVENTORY COST ANALYSIS

Improved accuracy reduces safety stock needs and optimizes replenishment. Using typical dairy parameters (25% annual holding cost, \$2.50 per unit stockout cost, and \$1.80 per unit waste cost), the annualized inventory cost impact is shown in Table 11.

SARIMAX reduces annual inventory costs by \$129,600 (15.4%) compared with \$91,500 (10.9%) for XGBoost. The incremental savings (\$38,100) highlight the financial justification for adopting SARIMAX.

2) WASTE REDUCTION ANALYSIS

Forecasting improvements reduce both overproduction waste and stockout-driven losses. Table 12 summarizes the projected reductions.

SARIMAX could reduce annual waste by 3,950 units (25.7%), compared with 2,420 units (15.7%) for XGBoost, equivalent to cost savings of \$7,110 vs. \$4,356 respectively.

3) SERVICE LEVEL IMPROVEMENTS

Enhanced accuracy reduces stockouts and improves availability. Table 13 shows service level impacts.

SARIMAX achieved a 96.6% average service level, exceeding the 95% target across all products, while XGBoost reached 94.2%, falling short in some categories. This 2.4 point gap translates into tangible customer satisfaction and competitive advantages.

4) SUMMARY

Across inventory, waste, and service dimensions, SARIMAX consistently delivers higher economic and operational benefits. These findings underscore the practical relevance of adopting advanced time-series approaches for perishable goods supply chains.

VI. DISCUSSION

A. THEORETICAL IMPLICATIONS AND CONTRIBUTIONS

This study contributes to the theoretical understanding of demand forecasting by demonstrating the conditions under which traditional statistical models can outperform advanced machine learning approaches. The superior performance of SARIMAX models challenges prevailing assumptions of machine learning dominance and highlights the importance of aligning model choice with data characteristics [49], [50].

The advantage of SARIMAX rests on three principles: (1) strong seasonal demand cycles in dairy products are well captured by the Box–Jenkins framework; (2) the integration of exogenous variables (e.g., weather, holidays, economic indicators) provides theoretical rigor while maintaining interpretability [51]; and (3) SARIMAX models exhibit a favorable bias–variance tradeoff compared to highly flexible but variance-prone machine learning methods such as XGBoost [31]. These findings support ensemble theory arguments that optimal model selection should depend on time series characteristics rather than universal application of any single method [52]. It is important to note that while point estimates favored SARIMAX for Zabadi, the bootstrap confidence interval included zero (Table 8), indicating that

TABLE 11. Annual inventory cost impact analysis.

Product	Current Cost	SARIMAX Cost	XGBoost Cost	SARIMAX Savings	XGBoost Savings
Laban	$145,600	$123,400	$131,200	$22,200 (15.2%)	$14,400 (9.9%)
Shanenah	$78,900	$67,200	$71,800	$11,700 (14.8%)	$7,100 (9.0%)
Labneh	$189,300	$158,700	$172,400	$30,600 (16.2%)	$16,900 (8.9%)
Jameed	$34,500	$26,800	$31,200	$7,700 (22.3%)	$3,300 (9.6%)
Milk	$267,800	$228,900	$229,400	$38,900 (14.5%)	$38,400 (14.3%)
Zabadi	$123,700	$105,200	$112,300	$18,500 (15.0%)	$11,400 (9.2%)
Total	$839,800	$710,200	$748,300	$129,600 (15.4%)	$91,500 (10.9%)

TABLE 12. Annual waste reduction impact.

Product	Current	SARIMAX	XGBoost	SARIMAX Red.	XGBoost Red.
Laban	2,340	1,760	1,980	580 (24.8%)	360 (15.4%)
Shanenah	1,890	1,420	1,610	470 (24.9%)	280 (14.8%)
Labneh	3,120	2,280	2,650	840 (26.9%)	470 (15.1%)
Jameed	780	520	640	260 (33.3%)	140 (17.9%)
Milk	4,560	3,420	3,780	1,140 (25.0%)	780 (17.1%)
Zabadi	2,670	2,010	2,280	660 (24.7%)	390 (14.6%)
Total	15,360	11,410	12,940	3,950 (25.7%)	2,420 (15.7%)

TABLE 13. Service level impact analysis.

Product	Current	SARIMAX	XGBoost
Laban	91.2%	96.8%	94.7%
Shanenah	89.8%	96.5%	93.9%
Labneh	90.5%	95.9%	93.2%
Jameed	87.3%	97.2%	92.1%
Milk	93.1%	96.7%	96.5%
Zabadi	91.8%	96.4%	94.8%
Average	90.6%	96.6%	94.2%

the difference was not statistically significant. This nuance aligns with the results for Milk, where model selection was neutral, underscoring that optimal forecasting approaches are product-dependent.

B. PRACTICAL IMPLICATIONS FOR DAIRY MANUFACTURING

From an operational perspective, improved forecasting translates directly into financial and service-level benefits. This study demonstrates that SARIMAX can reduce inventory costs by 15–25% while increasing service levels beyond 95%, providing a compelling case for investment in advanced forecasting systems.

The ability to anticipate demand surges during cultural events such as Ramadan enables proactive production planning and raw material procurement [53]. Enhanced forecast accuracy reduces refrigeration-related holding costs, minimizes waste, and improves cold chain coordination. Furthermore, improved demand visibility enhances supplier alignment and distribution scheduling, yielding sustainability benefits through reduced waste and optimized logistics [54], [55]. These findings are particularly valuable for small and medium-sized dairy enterprises lacking resources for large-scale machine learning infrastructures.

C. METHODOLOGICAL CONTRIBUTIONS AND INNOVATIONS

This research advances forecasting methodology through several innovations. The integration of breakpoint detection (Bai–Perron) with SARIMAX provides a robust mechanism for identifying structural shifts in demand associated with cultural and seasonal events [29]. The incorporation of lunar calendar variables addresses forecasting challenges in markets where religious observances follow non-solar cycles, with applicability to food and beverage industries beyond dairy.

The systematic feature engineering framework—incorporating lagged demand, rolling statistics, weather data, and promotional activities—demonstrates how external information can enhance predictive performance. SHAP-based feature importance analysis adds interpretability to XGBoost outputs, aligning with the growing emphasis on explainable AI in operations management [44]. Finally, the rigorous validation protocol (walk-forward validation, Diebold–Mariano testing, bootstrap CIs, and business impact quantification) establishes a comprehensive template for evaluating forecasting models.

D. COMPARISON WITH EXISTING LITERATURE

The results confirm earlier research showing the continued relevance of traditional time series methods in strongly seasonal contexts [56]. However, the observed performance gap between SARIMAX and XGBoost exceeds prior reports, underscoring the role of cultural and religious demand drivers often overlooked in the forecasting literature.

The inclusion of economic and business impact analysis extends prior studies that focus narrowly on accuracy metrics. This study demonstrates that even modest improvements in forecast accuracy can yield significant financial and sustainability benefits [57]. Moreover, the findings

challenge literature advocating machine learning as universally superior [14], reinforcing contingency-based model selection strategies.

E. LIMITATIONS AND VALIDITY CONSIDERATIONS

Several limitations affect the generalizability of findings. First, the study relies on data from a single Middle Eastern dairy facility, limiting cross-regional external validity. Strong cultural influences—such as Ramadan—may not apply in other markets. Second, the one-year dataset captures full seasonal cycles but not long-term structural shifts or evolving consumer preferences. Third, the analysis focuses on six core dairy products; results may differ for niche or highly innovative items. Future research should validate results across multiple sites, longer horizons, and broader product portfolios.

F. FUTURE RESEARCH DIRECTIONS

Building on these findings, several research avenues are recommended. Multi-site validation across diverse geographies would enhance generalizability and inform adaptive model selection frameworks. Integration of IoT-based real-time data and edge computing architectures could improve responsiveness in Industry 4.0 contexts. Advanced deep learning architectures (e.g., LSTM, Transformers) merit exploration, particularly in hybrid frameworks that combine statistical interpretability with machine learning flexibility [58]. Finally, future work should integrate sustainability metrics into forecasting evaluation, extending the scope from cost savings to environmental impact reduction.

VII. CONCLUSION

A. SUMMARY OF KEY FINDINGS

This study provided a comprehensive comparison of SARIMAX and XGBoost models for dairy demand forecasting, using rigorous statistical testing and economic analysis. Results showed that SARIMAX consistently outperformed XGBoost for most product categories, achieving an average MAPE of 5.55% versus 7.04% for XGBoost. Though the 1.49 percentage point gap appears modest, it translated into substantial business value: a 15.4% reduction in inventory costs and a 25.7% reduction in waste. Diebold–Mariano testing and bootstrap confidence intervals confirmed the statistical significance of these results.

The findings emphasize that optimal model selection is product-specific. Products with strong annual seasonality and cultural demand spikes, such as Jameed, benefited most from SARIMAX, while stable-demand products (e.g., milk) showed narrower gaps. The integration of exogenous variables, particularly weather, holiday calendars, and economic indicators, was essential to enhancing forecast accuracy across both approaches.

B. THEORETICAL CONTRIBUTIONS

This research makes several contributions to forecasting theory. First, the systematic comparison framework extends evaluation beyond accuracy metrics to include statistical significance and economic outcomes. Second, the integration of breakpoint detection with SARIMAX introduces a novel way to address structural changes in demand. Third, the study highlights the bias–variance tradeoff: parsimonious, theory-driven SARIMAX models often outperform flexible machine learning methods in seasonal contexts. Finally, incorporating cultural and religious factors into forecasting models addresses a critical gap in the literature, offering a framework applicable to industries facing culturally driven demand variation.

C. PRACTICAL IMPLICATIONS AND RECOMMENDATIONS

For practitioners, the results highlight the operational and financial benefits of advanced forecasting. SARIMAX models provided superior accuracy while requiring less computational infrastructure, making them well-suited for small and medium-sized manufacturers.

D. IMPLEMENTATION RECOMMENDATIONS

1) Apply systematic model selection procedures tailored to product characteristics rather than universal adoption of one method. 2) Invest in data collection and integration, prioritizing weather, economic indicators, and cultural event calendars. 3) Emphasize interpretability and business understanding to support adoption, as SARIMAX models are more transparent than black-box ML methods. 4) Embed forecasting improvements within broader digital transformation initiatives covering production, inventory, and supply chain planning.

E. LIMITATIONS AND FUTURE RESEARCH

Several limitations constrain generalizability. The single-site dataset from a Middle Eastern facility reflects cultural consumption patterns (e.g., Ramadan) that may differ in other regions. The one-year horizon captures seasonal effects but not longer-term structural changes or evolving consumer preferences. Moreover, analysis focused on six core dairy products, limiting applicability to niche or innovative product categories.

Future research should extend validation to multiple sites, longer horizons, and diverse product portfolios. Integration of IoT-based real-time data, edge computing, and deep learning architectures (e.g., LSTM, Transformers) represents promising directions. Hybrid frameworks combining statistical rigor with machine learning flexibility could further enhance forecasting capabilities. Finally, expanding analysis to sustainability metrics (e.g., CO_2 reduction, food waste) would strengthen the alignment between forecasting improvements and environmental objectives.

F. FINAL RECOMMENDATIONS

This research shows that significant improvements in dairy demand forecasting are achievable through systematic methodology selection, feature integration, and rigorous validation. SARIMAX models remain particularly effective

for seasonal, culturally sensitive products, while machine learning approaches provide competitive alternatives for stable-demand items. By combining interpretability, business value, and statistical rigor, the framework developed here provides practical guidance for manufacturers seeking to optimize forecasting within digital transformation initiatives.

Beyond the dairy industry, this replicable framework contributes to the broader field of perishable goods forecasting by combining methodological rigor with actionable business insights. The integration of cultural and seasonal drivers ensures that the study has both theoretical significance and practical value for global supply chains.

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